Science shows partial untangling (4.65× → 1.03×), though without the accompanying dimensional collapse observed in Law. Code maintains low G/L ratios at both scales (1.12× → 1.10×), consistent with its already-compact Lattice structure.

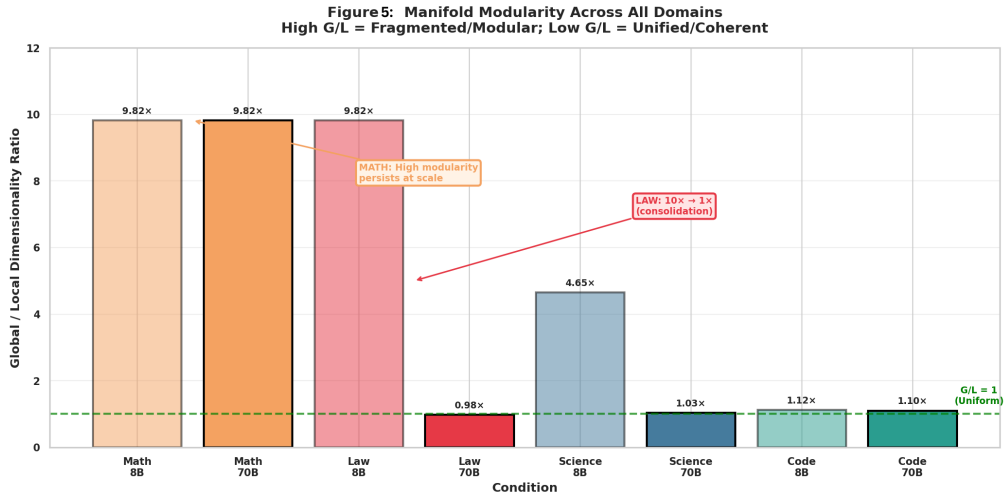

Figure 5. Manifold Modularity Across All Domains. Bar chart of Global/Local dimension ratio for all eight conditions. The green dashed line indicates $G/L = 1$ (uniform/flat manifold). Law exhibits 10× untangling (9.82× → 0.98×); Math remains highly modular (9.82×) at both scales; Science shows partial convergence; Code maintains low G/L throughout.

Figure 6 provides a simplified view of this phenomenon, directly comparing 8B and 70B G/L ratios for each domain. The red dashed line at ratio = 1 represents the theoretical optimum of a perfectly flat manifold. Law 70B is the only condition to reach this optimum, confirming that Crystallization involves complete geometric reorganization rather than incremental compression.

The radar chart in Figure 7 synthesizes the Crystallization signature for legal reasoning specifically. Comparing Law 8B (light red) to Law 70B (dark red) across four normalized metrics—Silhouette, Alignment, Compactness, and 1/G:L Ratio—reveals expansion along every axis. Law 70B dominates Law 8B on all metrics, forming a larger "fingerprint" that indicates comprehensive geometric restructuring rather than improvement on any single dimension.

Universal Dynamics: The Oscillatory Constant

Despite profound differences in topology across domains—Crystal, Liquid, and Lattice phases occupying distinct regions of the phase diagram—we discovered a universal constant in the dynamics of reasoning.

Figure 8 presents step-to-step trajectory coherence across all eight conditions. Coherence measures the cosine similarity between successive velocity vectors along the reasoning trajectory: +1 indicates smooth, unidirectional flow; 0 indicates random walk; −1 indicates perfect reversal at each step.

Remarkably, all conditions exhibit coherence of approximately −0.4, regardless of domain or scale. This negative coherence indicates a universal "zig-zag" pattern: the model does not move in a straight line toward the answer but oscillates, reversing direction partially at each step.

The universality of this constant is striking:

- **Law:** −0.40 (8B), −0.40 (70B)

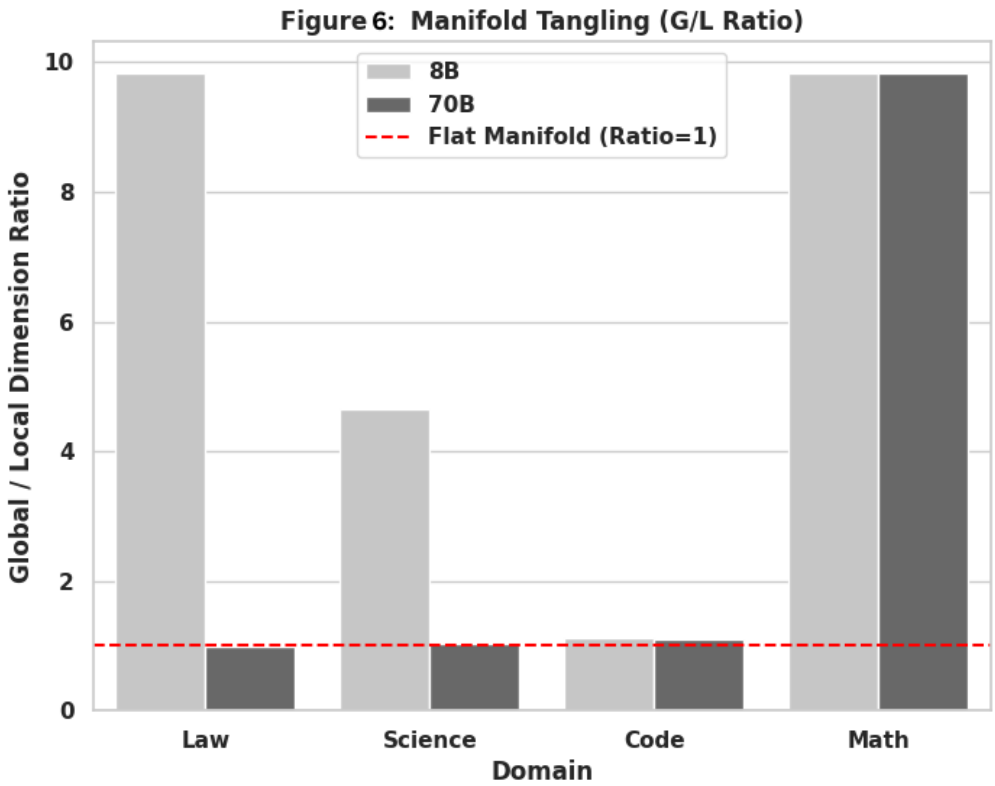

Figure 6. Manifold Tangling (G/L Ratio). Simplified comparison of 8B vs. 70B G/L ratios across domains. Law 70B uniquely achieves the flat-manifold optimum (ratio ≈ 1), while Math remains highly tangled at both scales.

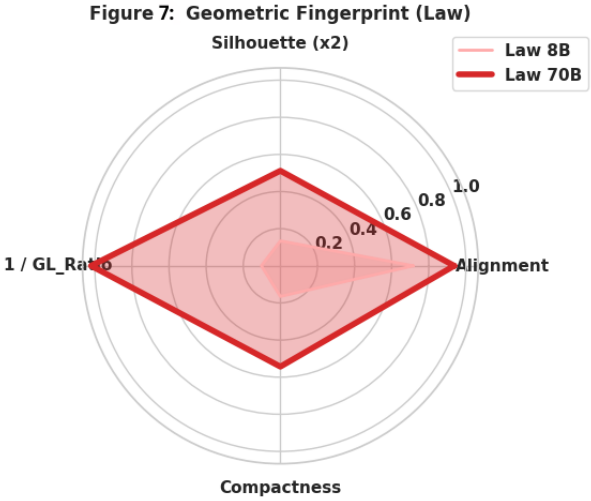

Figure 7. Geometric Fingerprint (Law). Radar chart comparing Law 8B vs. Law 70B across four geometric metrics. Law 70B (dark red) exhibits expansion along all axes, indicating comprehensive Crystallization: increased alignment, silhouette, compactness, and manifold coherence.

- **Science:** -0.40 (8B), -0.40 (70B)
- **Code:** -0.42 (8B), -0.40 (70B)
- **Math:** -0.42 (8B), -0.40 (70B)

Neither domain nor scale modulates this oscillatory signature. We interpret this as an architectural invariant of transformer dynamics—likely reflecting the interplay between attention (which aggregates context) and feedforward processing (which transforms representations). The specific value of -0.4 may encode the relative strengths of these competing processes in the Llama architecture.

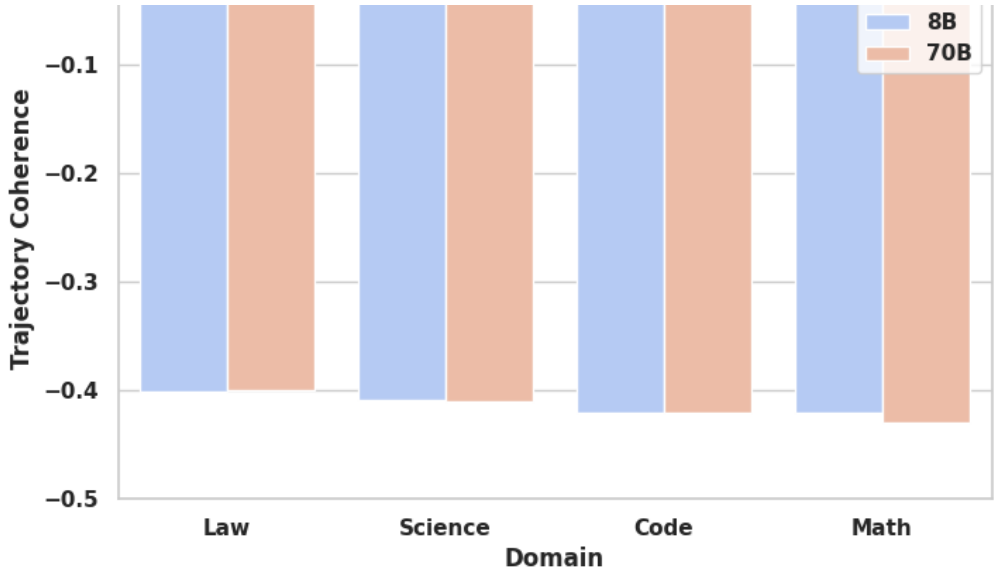

Figure 8. The Universal Oscillatory Constant. Bar chart of trajectory coherence across all conditions. All domains and scales exhibit coherence ≈ -0.4, indicating universal "zig-zag" dynamics. This negative coherence is robust to both domain (Crystal/Liquid/Lattice) and scale (8B/70B), suggesting it is a fundamental property of transformer reasoning.

Statistical Robustness

To ensure our findings reflect genuine geometric shifts rather than sampling noise, we computed effect sizes with 95% confidence intervals for the primary metrics.

Figure 9 presents the change in alignment ($\Delta\text{Alignment} = 70\text{B} - 8\text{B}$) for each domain with bootstrapped confidence intervals. Three findings emerge:

- **Law** shows the largest and most robust alignment gain: $\Delta = +0.22$ with a 95% CI of $[0.14, 0.30]$. The interval excludes zero, confirming statistical significance.
- **Science and Code** show moderate, significant gains: $\Delta = +0.15$ $[0.08, 0.22]$ and $\Delta = +0.18$ $[0.08, 0.28]$ respectively. These represent genuine improvements in directional coherence at scale, though without the accompanying dimensional collapse that characterizes Crystallization.
- **Math** shows no significant change: $\Delta = -0.01$ $[-0.07, +0.05]$. The confidence interval spans zero, confirming that mathematical reasoning exhibits true scale invariance in alignment as well as dimensionality.